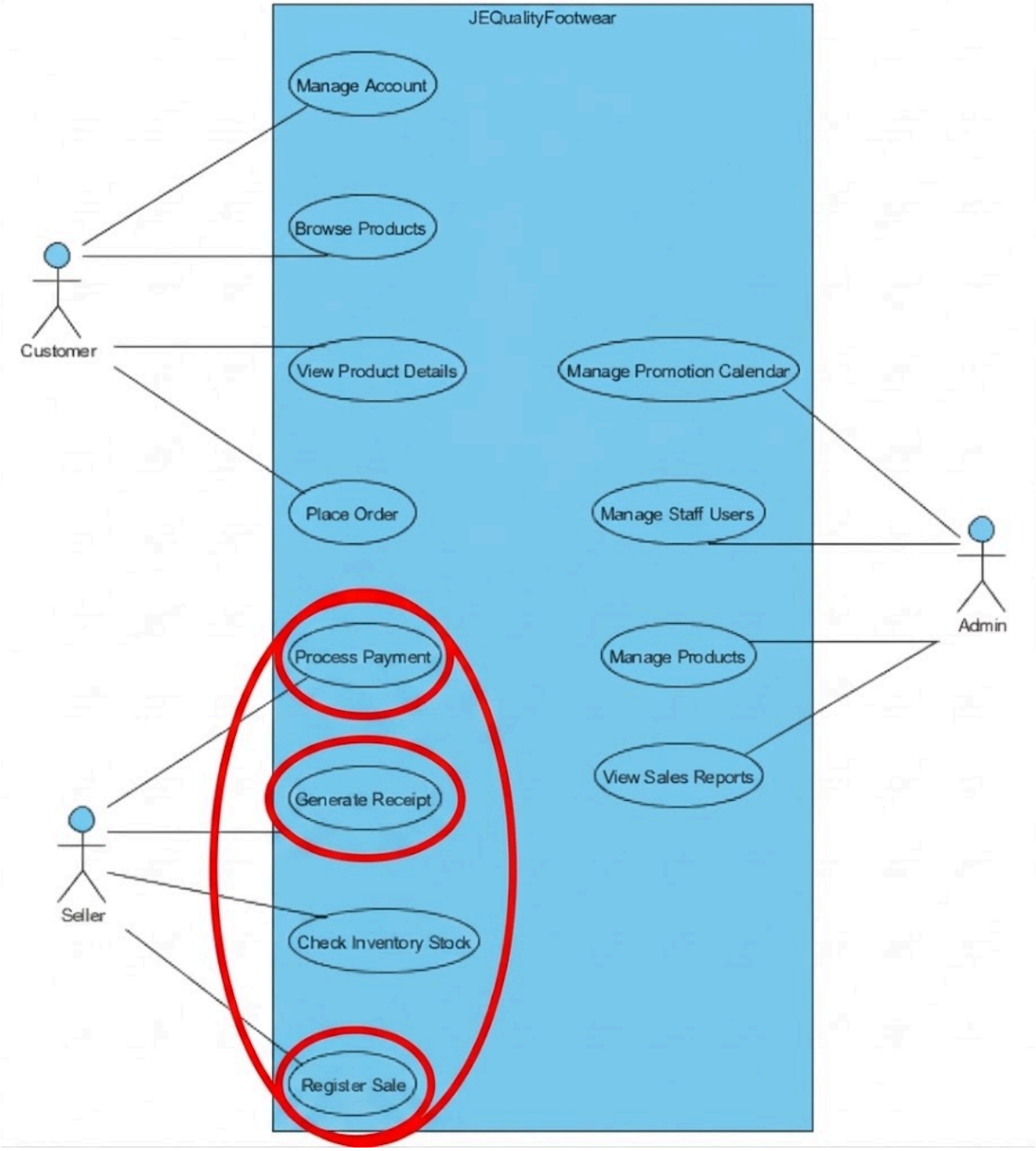
Inspector Team: Bytes & Fabric Devs
Inspected Team: Infinite Loops
Note: 10

USECASEDIAGRAM



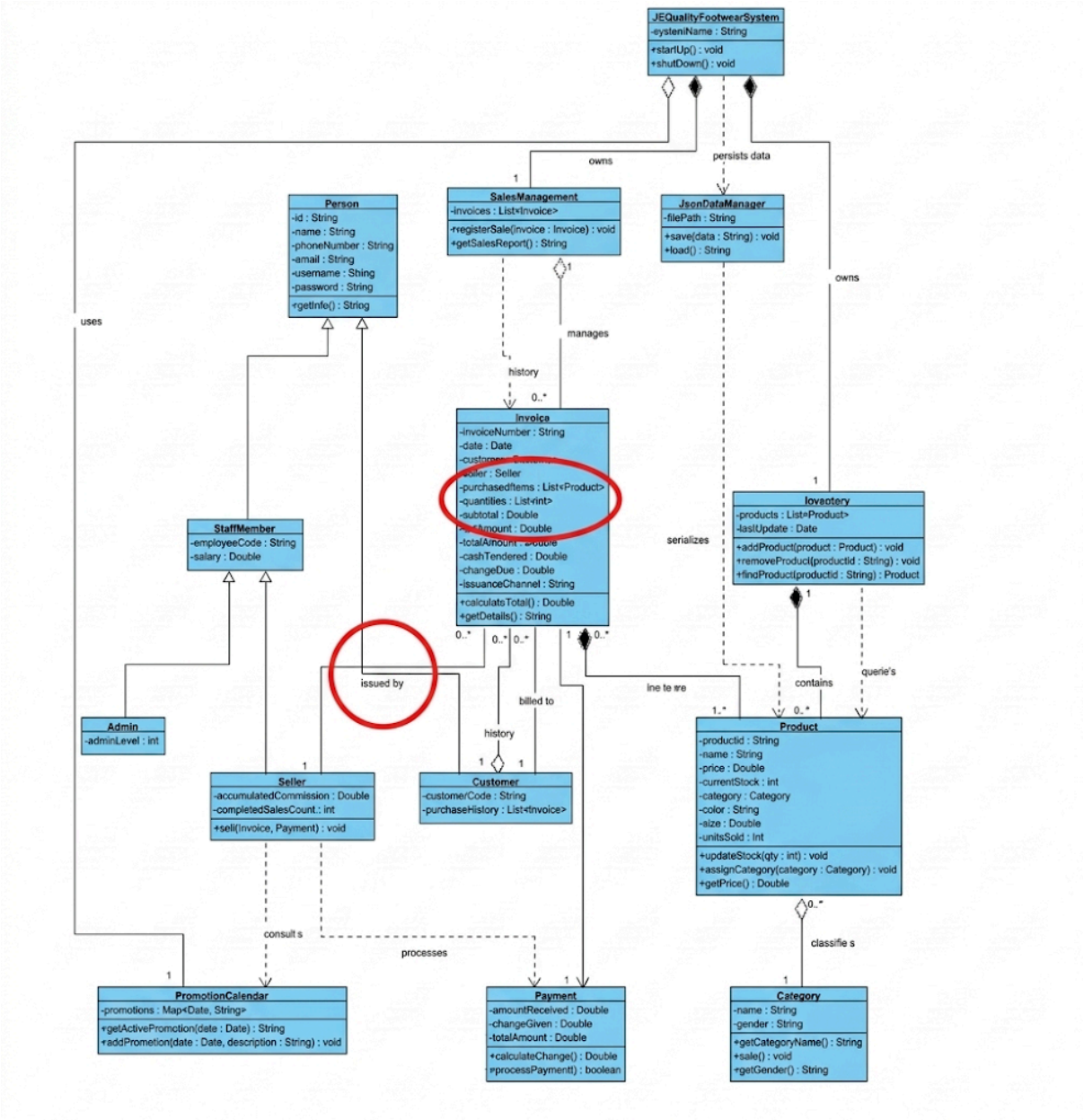
Identified Errors and Screenshot Location

- **Error 1 Orphan Use Case:** In the mid-left section of the system boundary, the Generate Receipt use case is completely disconnected, lacking any association or relationship with actors or other use cases.
- **Error 2 Missing Dependency Relationships :** The Process Payment use case is linked solely and directly to the Seller actor, missing the logical flow where payment processing derives from Place Order or Register Sale. Additionally, generating a receipt should be included within the payment process.

Proposed Solution

- Connect Generate Receipt using a dashed dependency line with the <<include>> stereotype originating from Process Payment or Register Sale.
- Structure the workflow correctly by adding <<include>> relationships from the primary processes (Place Order or Register Sale) to Process Payment to accurately reflect the core business sequence of the footwear system.

UML_JEQualityFootwear



Identified Errors and Screenshot Location

- The Error 1 : The Invoice class includes the internal attributes customer: Customer and seller: Seller, while simultaneously showing external association lines labeled billed to and issued by

pointing to Customer and Seller. Duplicating these links between internal attributes and diagram associations creates data inconsistency.

- The Error 2: The purchasedItems: List<Product> attribute inside Invoice maintains direct references to Product objects. If a product's price updates in the future, it will retroactively alter past invoice totals. Furthermore, maintaining a parallel quantities: List<int> list for item counts is prone to index misalignment errors.
- The Error 3: The diagram uses a composition relationship (black diamond) between Inventory and Product while simultaneously drawing a dashed association line labeled contains. Having two overlapping representations for the same link causes semantic ambiguity.

Proposed Solution

- Remove the internal attributes customer and seller from the Invoice class and rely solely on the billed to and issued by association lines along with their respective cardinalities .
- Introduce an intermediate association class (such as InvoiceLine) that stores the fixed quantity and exact unit price at the time of sale, decoupling historical invoices from future changes to the Product class.
- Retain only the standard UML aggregation/composition relationship between Inventory and Product and delete the redundant dashed contains association line.